GitHub + VS Code: Quick Setup (Windows & macOS)

Use the VS Code integrated terminal (PowerShell on Windows, Terminal/zsh on macOS). Git commands are the same on both systems.

1 Open your project folder and check Git

In VS Code: File > Open Folder, then Terminal > New Terminal. Check whether the folder is already a Git repository:

```
git status
```

2 Initialize only if needed

If Git reports “not a git repository,” initialize it. Otherwise skip this command.

```
git init
```

3 Add a .gitignore

Create **.gitignore** in the project root (beside the .git folder). Example Python/OpenCV exclusions: `__pycache__/`, `.venv/`, `data/train/`, `data/test/`, `results/`. Do not exclude source scripts.

4 Commit your files

Stage changes and save a local snapshot:

```
git add .
git commit -m "Initial project setup"
```

5 Connect to GitHub using HTTPS

On GitHub, open the repository and copy its **Code > HTTPS** URL. Use the first command if no remote exists; use the second if `origin` is already configured (including SSH):

```
git remote add origin https://github.com/USER/REPO.git
# OR, if origin already exists:
git remote set-url origin https://github.com/USER/REPO.git
```

6 Verify and push

Check the remote URL, then push your branch. Replace `main` with your branch name if different. For a repository with existing remote commits, pull/reconcile them before pushing.

```
git remote -v
git push -u origin main
```

7 Everyday collaboration

Pull updates before working; commit and push your changes afterward. Use feature branches and pull requests for team changes.

```
git pull
git add .
git commit -m "Describe changes"
git push
```

SSH authentication error? If you see `Permission denied (publickey)`, switch the remote from `git@github.com:...` to the HTTPS URL using `git remote set-url origin ...`. Git may prompt you to sign in via your browser.